

DS Assignment 1

Implement multi-threaded client/server Process communication using RMI.

//MyServer.java

```
import java.rmi.*;
import java.rmi.registry.*;

public class MyServer {
    public static void main(String args[]) {
        try {
            LocateRegistry.createRegistry(1099); // start RMI registry from code
            Concat stub = new ConcatRemote();
            Naming.rebind("rmi://localhost/concat", stub);
            System.out.println("Server ready and waiting...");
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

//MyClient.java

```
import java.util.Scanner;
import java.rmi.*;

public class MyClient {
    public static void main(String args[]) {
        try {
            Concat stub = (Concat) Naming.lookup("rmi://localhost/concat");
            Scanner input = new Scanner(System.in);
```

```

        String x, y;
        System.out.println("Enter String 1:");
        x = input.nextLine();
        System.out.println("Enter String 2:");
        y = input.nextLine();
        System.out.println("Concatenated Result: " + stub.concat(x, y));
    } catch (Exception e) {
        System.out.println("Error: " + e);
    }
}
}

```

//ConcatRemote.java

```

import java.rmi.*;
import java.rmi.server.*;

public class ConcatRemote extends UnicastRemoteObject implements Concat {
    ConcatRemote() throws RemoteException {
        super();
    }

    public String concat(String x, String y) {
        return x + y;
    }
}

```

//Concat.java

```

import java.rmi.*;

public interface Concat extends Remote {

```

```
    public String concat(String x, String y) throws RemoteException;  
}
```

Output:

Terminal 1:

```
javac *.java  
rmic ConcatRemote  
rmiregistry  
javac Concat.java ConcatRemote.java MyServer.java MyClient.java
```

Terminal 2:

```
java MyServer
```

Terminal 3:

```
java MyClient  
Enter String 1:  
KBT  
Enter String 2:  
COE  
Concatenated Result: KBTCOE
```